Math 1232 Summer 2025
Practice Final

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, **two-sided,** handwritten cheat sheet you've made for yourself ahead of time.
- You may not use a calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 12 problems, one on each mastery topic we've covered. You must do all of the problems on major topics, but need only complete 4 to 6 of the problems on secondary topics—your best 4 will count toward the exam grade, and up to an additional 2 problems can get you 1 or 2 bonus points.
- Each part of each topic is worth ten points. The whole test is scored out of 120 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

	a	b		a	b
M1:			M2:		
M3:			M4		
S1:		S2:		S3	
S4:		S5:		S6	
S7:		S8:		Total:	

Name:

Name:

Problem 1 (M1).

(a) Compute $\dfrac{d}{dt} 2^t \log_5(t)$.

(b) Compute $\displaystyle\int_0^{2^{-\frac{1}{3}}} \frac{x^2}{\sqrt{1-x^6}}\, dx$.

Name:

Problem 2 (M2). Compute the following integrals using any of the methods we have learned in class. Show enough work to justify your answers.

(a) $\displaystyle\int \frac{x^3}{\sqrt{x^2+4}}\,dx$

(b) $\displaystyle\int x^2 e^{3x}\,dx$

Name:

Problem 3 (M3). Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-2)^n}{n^3+n}$.

(b) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{5n^2+3n}$.

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Problem 4 (M4).

(a) Let $f(x) = \cos^2(x)$. Use *the definition of a Taylor series* to find $T_4(x, \pi)$ for this function. (That is, find the terms up through the degree four term of the series centered at $a = \pi$.)

(b) Write a power series expression for $\frac{x^4}{2-4x}$ centered a 0. What is the radius of convergence?

Name:

Problem 5 (S1). Let $f(x) = \sqrt[4]{x^7 + 5x^5 + 4x^3 + 6}$. Find $(f^{-1})'(2)$.

Problem 6 (S2). Compute $\lim_{x \to 0} \frac{\arctan(x)}{\ln(x+1)}$.

Name:

Problem 7 (S3). Compute $\int_1^5 \frac{1}{\sqrt{x-1}}\,dx$.

Problem 8 (S4). Compute the area of the surface obtained by taking the curve $x^{2/3}+y^{2/3}=1$ as x goes from 0 to 1 and rotating it around the x-axis.

Name: ____

Problem 9 (S5). Find the (particular) solution to the initial value problem given by $-2x + 4y^3\sqrt{x^2+3} \cdot y' = 0$ and $y(1) = 1$.

Problem 10 (S6). Compute $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n+4}{n+3}\right)$.

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Problem 11 (S7). Find the radius and interval of convergence of the series $\sum_{n=1}^{\infty} \frac{(5x-3)^n}{\sqrt{n}}$.

Problem 12 (S8). Find a (non-parametric) equation of the line tangent to the curve parametrized by $x = \cos^3(t), y = \sin^3(t)$ at the point $(1/8, -3\sqrt{3}/8)$.